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REVIEW

A Review of locomotor diversity in mammals with analyses exploring the influence of substrate use, body mass and intermembral index in primates

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arboreality; locomotor repertoire; positional behavior; shannon–wiener diversity index; phylogenetic analysis.

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Abstract

Locomotor diversity has meant many different things depending on the subject area or investigator. While no concrete definition of locomotor diversity is currently available, this has not stopped researchers from making a number of assertions about the underlying mechanisms that determine whether a species will have high or low locomotor diversity. Some of these claims include that: (1) arboreal primates demonstrate higher locomotor diversity than terrestrial species; (2) anatomically generalized and small-bodied primates demonstrate higher locomotor diversity than anatomically specialized and large-bodied taxa; and (3) primates demonstrate higher locomotor diversity compared to non-primate mammals. None of these claims have been tested in any sort of comparative framework. Using previously published locomotor repertoire data from 110 mammalian species, this study co-opted the Shannon–Wiener diversity index to calculate a singular measure of locomotor diversity. Phylogenetic analyses of these locomotor diversity indices reveal that within primates, small-bodied species demonstrate greater locomotor diversity than large-bodied taxa. However, this effect becomes non-significant when accounting for differences in substrate-use (i.e. arboreal vs. terrestrial). Anatomical specialization has no effect on locomotor diversity in primates. Furthermore, primates do not inherently have greater locomotor diversity than other mammalian taxa. Findings from this study suggest that movement on arboreal substrates requires all mammals, regardless of taxonomic affiliation, to demonstrate high locomotor diversity. These data further highlight the challenges faced by arboreal animals and raise the possibility that being able to switch fluidly and frequently between many different locomotor modes may be advantageous for animals moving in an arboreal milieu.

Introduction

A working definition of locomotor diversity is troublesome and means many different things depending on the specific field or investigator (Van Valkenburgh, 1985; Gebo, 1987; Gatesy & Middleton, 1997). For some, locomotor diversity refers to an ability of an animal to use many different aspects of the environment (e.g. high, middle and low canopy) (Fleagle & Mittermeier, 1980; Schmitt, Zeininger & Granatosky, 2016) or has a morphotype specific to a particular environment (Jones, 2003; Swartz, Freeman & Stockwell, 2003). More often than not, locomotor diversity refers to the number of distinct locomotor behaviors observed in a particular taxon (Lovejoy, 1978; Van Valkenburgh, 1985; Gebo, 1987; Schmitt *et al.*, 2016). However, this definition can be troublesome as it is: (1) dependent on the taxonomic level under investigation (e.g. a mammalian family should demonstrate a greater number of locomotor behaviors than a species within said family); (2) highly influenced by the investigator's initial ethogram and whether the investigator elects to split or lump locomotor behaviors in broad or narrow categories; and (3) overemphasizes the importance of uncommon or rare locomotor behaviors. This study takes a different approach by considering diversity in a way commonly used in the ecological literature (e.g. Pielou, 1966; Risser & Rice, 1971; Spellerberg & Fedor, 2003; Keylock, 2005). Therefore, from here on locomotor diversity will refer to the number locomotor behaviors and proportion of use observed *within* a species. Currently, there are no means of calculating locomotor diversity in this manner.

It has often been asserted that primates display the highest locomotor diversity compared to any other mammalian group (e.g. Vilensky & Larson, 1989; Cant, 1992; Blanchard & Crompton, 2011; Fleagle, 2013; Schmitt *et al.*, 2016). This high degree of locomotor diversity observed in primates has been considered an essential adaptation potentially responsible for the evolutionary success of the order (Jones, 1916; Vilensky & Larson, 1989; Schmitt *et al.*, 2016), and it is this assumption that has led to numerous studies documenting locomotor behaviors across a broad phylogenetic spectrum of primate species (Table S1 and Data S1). These locomotor profiles have provided essential information about behavioral ecology, functional morphology, paleontology and by extension, important data concerning primate locomotor evolution (e.g. Prost, 1965; Hunt *et al.*, 1996; Rein, Harvati & Harrison, 2015; Fabre *et al.*, 2017).

There has been extensive speculation as to what factors of primate biology might have led to the considerable diversification of locomotor modes (Vilensky & Larson, 1989; Cant, 1992; Schmitt *et al.*, 2016). Perhaps, the most cited explanation is the link to a long, and potentially restricted, evolutionary history tied to an arboreal environment in early primates (Jones, 1916; Schmitt *et al.*, 2016). The arboreal milieu is characterized by considerable variation in substrate diameter, orientation, compliance and continuity (Jones, 1916; Larson *et al.*, 2000). The complexity of this environment is thought to have put strong selective pressures on the ability of primates to rapidly switch between disparate locomotor behaviors to effectively (i.e. lower risk of injury and/or overall metabolic costs) adjust to substrate variation (Cant, 1992; Schmitt *et al.*, 2016). In contrast, terrestrial environments are commonly considered less complex, and animals within these environments likely have to deal with lower substrate variation (Jenkins, 1974). The greater substrate homogeneity observed in terrestrial landscapes is thought to relax the need to switch between different forms of locomotion in order to successfully to traverse the environment (Jones, 1916; Jenkins, 1974).

Another explanation for the greater locomotor diversity in primates comes from inherent features of primate anatomy. In general, primates retain a primitive and generalized post-cranial skeleton with grasping hands and feet, and highly mobile limb joints (Jones, 1916; Napier, 1967; Vilensky & Larson, 1989; Schmitt *et al.*, 2016). Together these features allow most primates greater flexibility in overall range of motion, and therefore capable of proficient locomotion in a number of environmental contexts (Jones, 1916; Larson *et al.*, 2000; Schmitt *et al.*, 2016). However, there are numerous primate species that have evolved anatomical specializations away from the primitive bauplän in order to better fit a particular form of locomotion (e.g. vertical-clinging and leaping or suspensory behavior), or environmental context (Prost, 1965; Napier, 1967; Hunt *et al.*, 1996; Fleagle, 2013). For these species, it has been suggested that overall locomotor diversity is lower when compared to more anatomically generalized species (Fleagle, 1978; Fleagle & Mittermeier, 1980; Gebo, 1987; Cant, 1992). Body mass also has been argued to influence the locomotor behaviors of primates. Specifically, smaller species/individuals are considered to use a greater range of locomotor behaviors than larger species/individuals that may be restricted in the locomotor modes they are able to adopt due to their larger body mass (e.g. orangutans are unlikely to leap) (Fleagle & Mittermeier, 1980; Cant, 1987, 1992). The relationship between locomotor diversity and anatomical specialization or body mass has not been tested in any sort of comparative framework.

A common assumption among those that study primate behavior and movement is that primates display the highest locomotor diversity compared to any other mammalian group (e.g. Vilensky & Larson, 1989; Cant, 1992; Blanchard & Crompton, 2011; Fleagle, 2013; Schmitt *et al.*, 2016). However, a direct comparison between the locomotor diversity of primates and non-primate mammals has never been made. Indeed, the range of locomotor behaviors observed in non-primate mammalian groups is quite impressive (Brown & Yalden, 1973; Van Valkenburgh, 1985; Samuels & Van Valkenburgh, 2008), and there are many locomotor behaviors used by other mammals that are not present in living primates (e.g. swimming, powered flight, burrowing). Sadly, opinions about locomotor diversity in primates and other mammals are likely primarily influenced by taxon-centric thinking and sampling bias. This concern is made blatantly clear by the results of the initial literature review for this study (Table S1 and Data S1). Current taxonomy suggests that there are approximately 230–250 species of living primate. Of the 174 total locomotor repertoires used in this study, 157 of these were collected from primates. In contrast, the locomotor repertoires of only 17 non-primate species were available from the remaining approximately 5000 mammalian species. Therefore, a review of locomotor behavior and diversity across primates and other mammals is sorely needed.

This study aims to fill this gap by conducting a broad comparative analysis of mammalian locomotor diversity and providing a concrete and comparable definition of locomotor diversity. To address these aims, this study utilizes diversity indices, which are commonly utilized in the ecology literature, to summarize and consolidate locomotor behavior into a single comparable value. With these locomotor diversity indices, this study aims to address the following hypotheses:

H1: Locomotor diversity is influenced by phylogenetic relatedness (i.e. more closely related species will demonstrate greater similarities in locomotor diversity than more distantly related taxa).
H2: Arboreal primates demonstrate higher locomotor diversity indices than terrestrial primates.
H3: Anatomically generalized and small-bodied primates demonstrate higher locomotor diversity indices than anatomically specialized and large-bodied primates.
H4: Primates demonstrate higher locomotor diversity indices compared to non-primate mammals.

Materials and methods

Data collection

It was first necessary to acquire the locomotor repertoires from a range of mammalian species. These data were assembled by searching previously published material in academic search engines. The keywords used to find these data included: 'Positional Behavior', 'Positional repertoire', 'Locomotor Behavior', 'Locomotor repertoire', 'Locomotor diversity', 'Habitat use', 'Range use'. The search engines used included Academic Search, BioOne, Google Scholar, Microsoft Academic, PubMed and Web of Science. Searches were confined to papers published before August 2018. All data were collected from the locomotor behavior of adults, and locomotor behavior from multiple substrates or sexes were pooled for analysis. Care was taken to assure that the names of the locomotor behaviors were not modified from those defined in the original source (Table S1 and Data S1).

To calculate locomotor diversity indices (described below), all data must equal 100% of the total locomotor repertoire. In most instances, authors provided this information as needed, but in some cases the frequency of locomotor behavior was presented as a percentage of overall positional behavior. In these instances, all data were converted into a percentage of the total locomotor behavior. It was also the case that some papers reported the frequencies of locomotor behaviors on figures, rather than tables or in text. In these cases, data were extracted using DataThief III (Tummers, 2016), which has been shown to be a reliable and repeatable data extraction tool (Flower, McKenna & Upreti, 2016).

Calculating the locomotor diversity index

This study co-opted the Shannon–Wiener diversity index to calculate a singular measure of locomotor diversity. The Shannon–Wiener diversity index is traditionally used in the ecology literature to provide a statistically comparable metric to quantify the diversity of species composition in a specific area of interest (e.g. forest, transect, etc.) (Pielou, 1966; Risser & Rice, 1971; Spellerberg & Fedor, 2003; Keylock, 2005). This study employs this same logic using the proportion of each locomotor behavior observed within a species to calculate a species-specific locomotor diversity index. The locomotor diversity index is calculated as:

$$\text{Locomotor diversity index} = -\sum p_{\mathrm{i}} \ln p_i$$

where p_{i} is the proportion of a particular locomotor behavior (e.g. quadrupedal walking, leaping, climbing) out of the total combination of locomotor behaviors (e.g. proportion of quadrupedal walking + proportion of leaping + proportion of climbing) (Fig. 1). A low locomotor diversity index indicates that a particular species frequently uses only few different locomotor behaviors. High locomotor diversity index values indicate that a particular species frequently uses many different locomotor behaviors.

Statistical analyses

To test for the influence of phylogeny on locomotor diversity, both Blomberg's K (Blomberg, Garland & Ives, 2003) and Pagel's λ (Pagel, 1999) were calculated. These values were assessed for the entire sample pooled, only primate data and only non-primate mammal data.

A Mann–Whitney U-test and a phylogenetic ANOVA (Garland *et al.*, 1993; Revell, 2012; Rohlfs & Nielsen, 2015) were used to test for differences in locomotor diversity between primates and non-primate mammals and between arboreal and terrestrial primates. Arboreality versus terrestrially in primates was based on descriptions in originally published manuscripts (Table S1 and Data S1) and Fleagle (2013). An overall locomotor diversity index was calculated for all primates and all non-primate mammals and for all arboreal and all terrestrial primates. As large differences in sample size between the groups could artificially inflate or deflate an overall raw locomotor diversity index, this study co-opted Pielou's evenness index (Pielou, 1966), which was calculated simply by dividing the locomotor diversity index by the natural logarithm of the total number of species in each group. A z-score was calculated to test the statistical probability that the overall locomotor diversity index and Pielou's evenness index for all primates and all non-primate mammals and all arboreal and all terrestrial primates were sampled from the same normal distribution (Pocock, 2006). An additional non-phylogenetic Mann–Whitney U-test was used to assess the potential effects of observational data collected from animals in a captive versus wild environment.

To test for the influence of anatomy on locomotor diversity index in primates, this study used intermembral index (IMI) as a proxy anatomical specialization. This metric is calculated as the length of the forelimbs (humerus + radius) divided by the length of the hindlimbs (femur + tibia) multiplied by 100. The IMI is used frequently in primatology, since it helps predict primate locomotor patterns. For scores lower than 100, the forelimbs are shorter than the hindlimbs, which are common for anatomically specialized vertical clinging and leaping species. Anatomically generalized species have scores around 100, while arm-swinging primates have scores significantly higher than 100. A phylogenetic least squares regression (PGLS) was used to assess the relationship between locomotor diversity index and IMI (Garland, Harvey & Ives, 1992; Revell, 2012; Symonds & Blomberg, 2014). A second PGLS was used to test the association between locomotor diversity index and body mass in primates. A species mean body mass was calculated from the average mass of both sexes. All body mass data were collected from Fleagle (2013). Only phylogenetic regressions were used because, while it is unclear whether locomotor diversity index is influenced by phylogenetic-relatedness, there is a strong and significant phylogenetic signal for both body mass (Blomberg's $K = 2.29$, $P \leq 0.001$; Pagel's $\lambda = 1.00$, $P \leq 0.001$) and IMI (Blomberg's $K = 2.16$, $P \leq 0.001$; Pagel's $\lambda = 1.00$, $P \leq 0.001$) in primates.

Prior to any analyses, locomotor diversity indices, body mass and IMI were log-transformed. All phylogeny-based analyses were performed in the statistical packages phytools (Revell, 2012), ape (Paradis, Claude & Strimmer, 2004) and geiger (Harmon *et al.*, 2007). For all phylogenetic analyses, the phylogeny was constructed by pruning a recent supertree (Hedges *et al.*, 2015) to include only the species in our study. In the case that multiple studies reported locomotor behavioral data from the same species, a species mean locomotor diversity index was used for all phylogenetic analyses. All non-

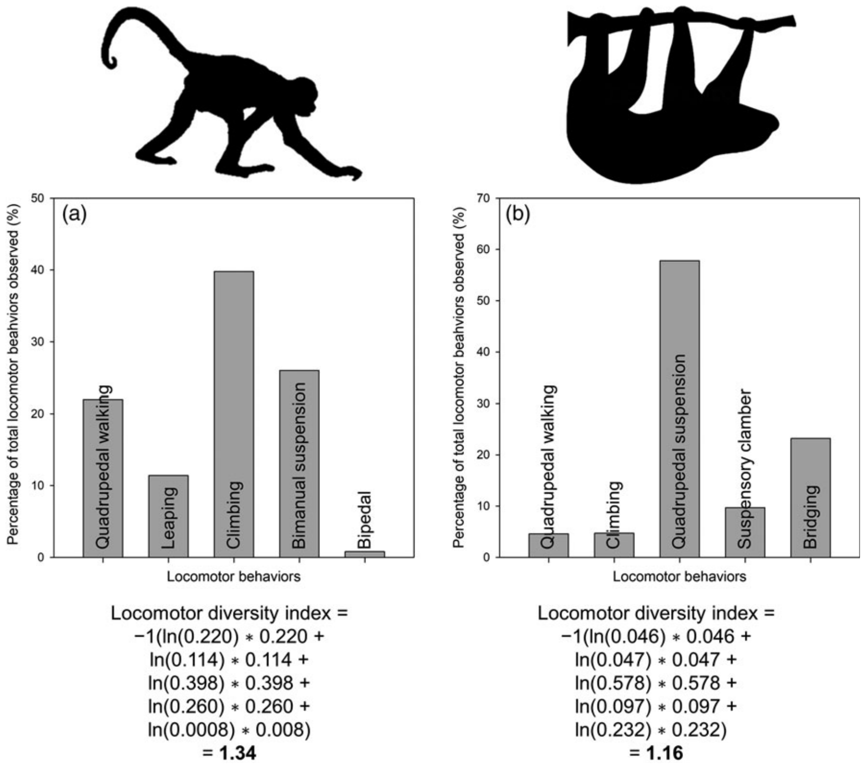

Figure 1 An example of how to calculate the locomotor diversity index for a (a) primate (i.e. *Ateles geoffroyi*) and (b) non-primate mammal (i.e. *Choloepus didactylus*). The locomotor diversity index is calculated as: $-\sum p_i \ln p_i$, where p_i is the proportion of a particular locomotor behavior out of the total combination of locomotor behaviors.

phylogeny-based analyses were performed in MATLAB (v.2017b; MathWorks, Natick, MA, USA).

Results

In total, locomotor frequency data were available from 174 studies consisting of 110 species (93 primates and 17 non-primate mammals; Table 1). Across mammals, quadrupedal walking is the most common locomotor mode representing ~36% of the locomotor behaviors observed. Other commonly used forms of locomotion include leaping (~23%) and climbing (~15%) (Fig. 2). Significant differences in locomotor diversity indices were observed ($P = 0.027$) between data collected from species in a captive ($n = 39$, 1.26 ± 0.24) versus wild ($n = 126$, 1.10 ± 0.41) setting (Table S1).

Overall, the phylogenetic signal in locomotor diversity index across mammals was low and not significant (Blomberg's $K = 0.08$, $P = 0.353$; Pagel's $\lambda < 0.001$, $P = 1.00$) (Fig. 3). This finding was similar for the analysis of just the primate data (Blomberg's $K = 0.14$, $P = 0.185$; Pagel's $\lambda = 0.03$, $P = 0.69$) and just the non-primate mammal data (Blomberg's $K = 0.26$, $P = 0.546$; Pagel's $\lambda < 0.001$, $P = 1.00$).

Within primates, species classified as being primarily arboreal ($n = 84$, 1.22 ± 0.31) had significantly greater (Mann–Whitney U-test: $P = 0.001$; Phylogenetic ANOVA: $P = 0.001$) locomotor diversity indices than primarily terrestrial ($n = 9$, 0.69 ± 0.43) taxa (Fig. 4a). However, no significant differences were observed between the overall locomotor diversity index (arboreal = 2.19 vs. terrestrial = 0.34; $z = 1.16$, $P = 0.122$) or Pielou's evenness index (arboreal = 1.14 vs. terrestrial = 0.18; $z = 0.64$, $P = 0.260$) between arboreal and terrestrial primates. A significant negative ($y = -25x + 0.78$; $R^2 = 0.085$) relationship was observed between body mass and locomotor diversity index ($P = 0.008$) in primates, but this relationship becomes non-significant ($P = 0.282$) when accounting for substrate (arboreal vs. terrestrial) as an additional predictor. No significant relationship ($P = 0.460$) between locomotor diversity index and IMI was observed (Fig. 5).

No significant difference (Mann–Whitney U-test: $P = 0.194$; Phylogenetic ANOVA: $P = 0.755$) in locomotor diversity indices were observed between primate ($n = 93$, 1.17 ± 0.36) and non-primate mammals ($n = 17$, 1.29 ± 0.25) (Fig. 4b). Additionally, no significant differences were observed between the overall locomotor diversity index (primates = 1.92 vs. non-primate mammals = 2.29; $z = 0.18$, $P = 0.429$) or Pielou's evenness index (primates = 0.98 vs. non-primate mammals = 1.86; $z = 0.53$, $P = 0.300$) between primates and non-primate mammals.

Discussion

The definition and calculation of locomotor diversity as used in this study provides a repeatable and comparable means of describing how an animal navigates its environment that is far more informative than simply a raw count. The observation of

Table 1 Taxonomic breakdown of the sample used in this study. See Table S1 and Data S1 for original source material

Order	Family	Number of species in each Family
Didelphimorphia	Caluromyidae	1
Diprotodontia	Acrobatidae	1
Pilosa	Megalonychidae	1
Carnivora	Mustelidae	1
Rodentia	Muridae	1
	Sciuridae	9
Scandentia	Tupaiidae	2
Dermoptera	Cynocephalidae	1
	Galagidae	4
	Lorisidae	3
	Daubentoniidae	1
	Cheirogaleidae	4
	Lemuridae	8
	Lepilemuridae	1
	Indriidae	3
Primates	Tarsiidae	4
	Callitrichidae	8
	Pitheciidae	5
	Cebidae	6
	Atelidae	8
	Cercopithecidae	28
	Hylobatidae	5
	Hominidae	5

a relatively high locomotor diversity index in a species indicates that said species uses many different locomotor behaviors frequently as it moves through its environment. While raw counts of locomotor behaviors may provide similar interpretation, they are inherently faced with issues. As a thought experiment, consider two species. Species A demonstrates four distinct locomotor behaviors and uses them all equally. In contrast, Species B uses 10 distinct locomotor behaviors, but uses one locomotor mode 90% of the time and the remaining 10% of the locomotor repertoire is split evenly across the remaining nine behaviors. If one was to use only raw counts, Species B would be considered to have greater locomotor diversity, but this interpretation is obviously incorrect. In turn, the use of the locomotor diversity index reveals that Species A (Locomotor diversity index = 1.39) demonstrates a more varied locomotor repertoire compared to Species B (Locomotor diversity index = 0.54), which should be considered a locomotor specialist. It should be noted, that although not as influenced as a raw count, the calculation of a locomotor diversity index is still affected by the researcher's original ethogram. This limitation is especially prudent when the researcher uses limited locomotor categories and shifts a large proportion of the locomotor behaviors observed to an 'Other' category. This study suggests that researcher's interested in quantifying the positional behaviors of animals should utilize Hunt *et al.* (1996) as a guideline in the development of ethograms.

There is very little influence of phylogeny on patterns of locomotor diversity across mammals. This means that closely related species are not more likely to have similar levels of locomotor diversity compared to more distantly related taxa. This finding is not overly surprising, as behavioral characteristics are well-known to be more evolutionarily labile compared to other traits (Blomberg *et al.*, 2003). Related to this, and contrary to the original hypothesis, this study found no support for the assumption that primates demonstrate greater levels of locomotor diversity compared to other mammalian groups. This finding provides direct contradictory evidence to a long-held idea among primatologists that is often discussed in introductory anthropology courses and books (Fleagle, 2013). The number of studies dedicated to describing and quantifying the positional behavior of primates further propagates the 'elite' status of primates in the realm of animal locomotion (Table S1 and Data S1). Although the aim of this study is not to criticize these fields of endeavor, it is possible that this fierce primatocentric thinking about locomotor diversity may have had detrimental effects on constructive thoughts about locomotor behaviors of non-primate mammals. This speaks to the necessity of collecting positional behavior profiles from non-primate species, as it is likely that we have yet to capture the full range of locomotor abilities used by mammals. That having been said, the limited sample of non-primate mammals represents a major limitation of this study. This small sample likely does not capture the range of variation in locomotor modalities observed across mammals. Furthermore, there is a bias for data collected from arboreal species that likely overestimates locomotor diversity among non-primate mammals (e.g. locomotor diversity in ungulates is likely low due to primarily terrestrial movement).

Instead of some kind of hard and fast taxonomic break that separates primates from other mammals, it is likely that the primary determinate of locomotor diversity is substrate-use. Specifically, by locomoting in an arboreal niche, animals are faced with a complex three-dimensional environment with discontinuous supports of varying diameter, inclination and compliance. To successfully move and forage through this environment, animals must use a range of locomotor behaviors. This idea is emphasized by looking at the non-primate mammals analyzed in this study. These taxa are primarily arboreal, and as such the comparisons with primates (i.e. a primarily arboreal clade) revealed no significant differences. A closer look at the great apes further stresses this point by contrasting the levels of locomotor diversity observed in the highly arboreal *Pongo sp.* and *Pan paniscus* compared to the more terrestrial *Gorilla gorilla* and *Pan troglodytes*. The same stark differences in locomotor diversity can be seen between the primarily arboreal (e.g. *Cercopithecus* and colobines) and terrestrial (e.g. *Papio* and *Erythrocebus*) Old World monkeys (Fig. 3). Taken together, it's likely that primates demonstrate high levels of locomotor diversity not because of some inherent neuromuscular flexibility that better allows them to adopt and utilize novel locomotor modalities (Vilensky & Larson, 1989; Schmitt *et al.*, 2016), but instead because the majority of the lineage is arboreal. A more detailed look at other non-primate arboreal species (e.g. opossums, kinkajous, fossa) would most likely support this assertion. It should be noted that the arboreal versus terrestrial dichotomy observed in this study may be influenced by the large disparity in arboreal (84

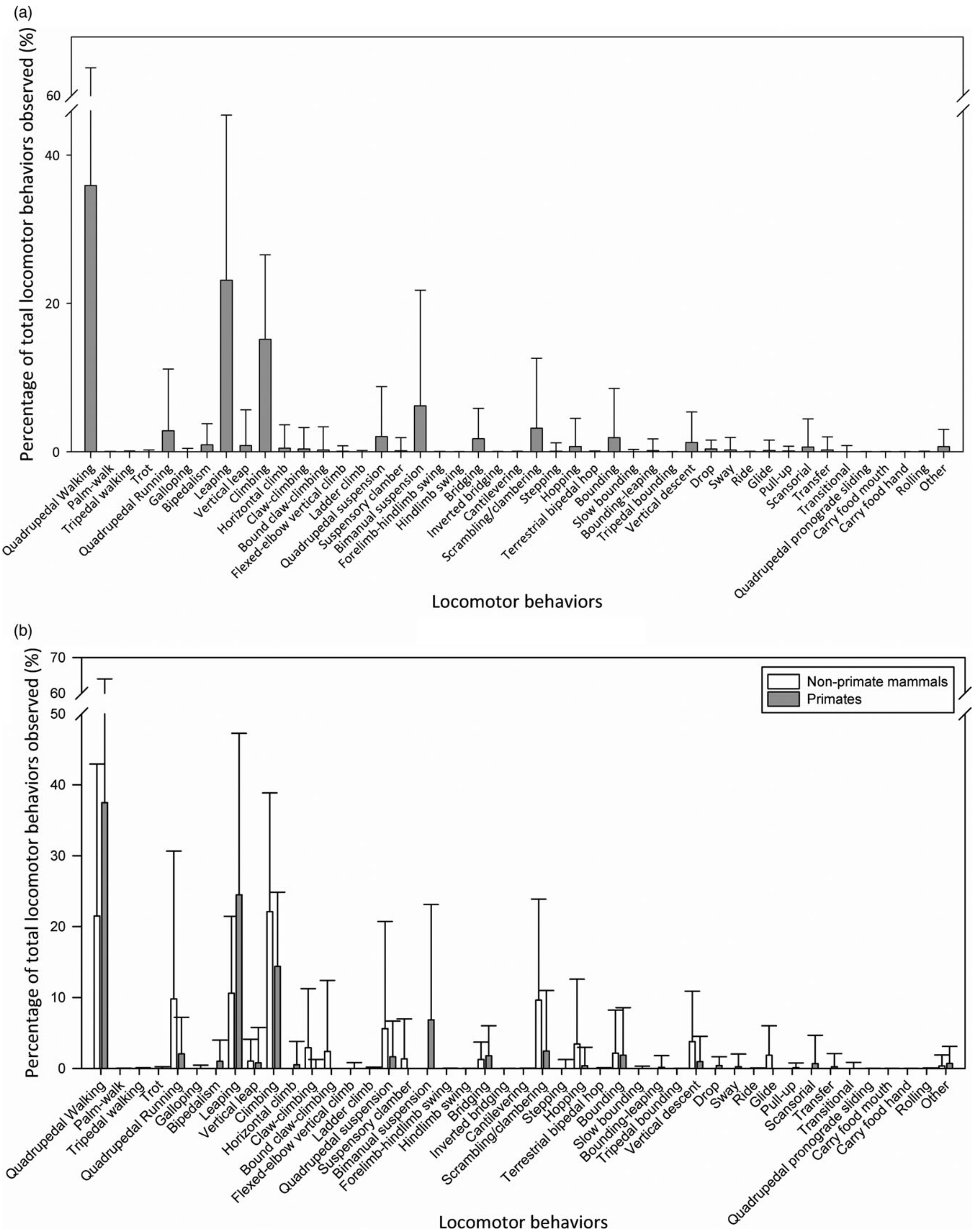

Figure 2 The percentages of locomotor behaviors observed for (a) all species in this study and (b) primates and non-primate mammals separately.

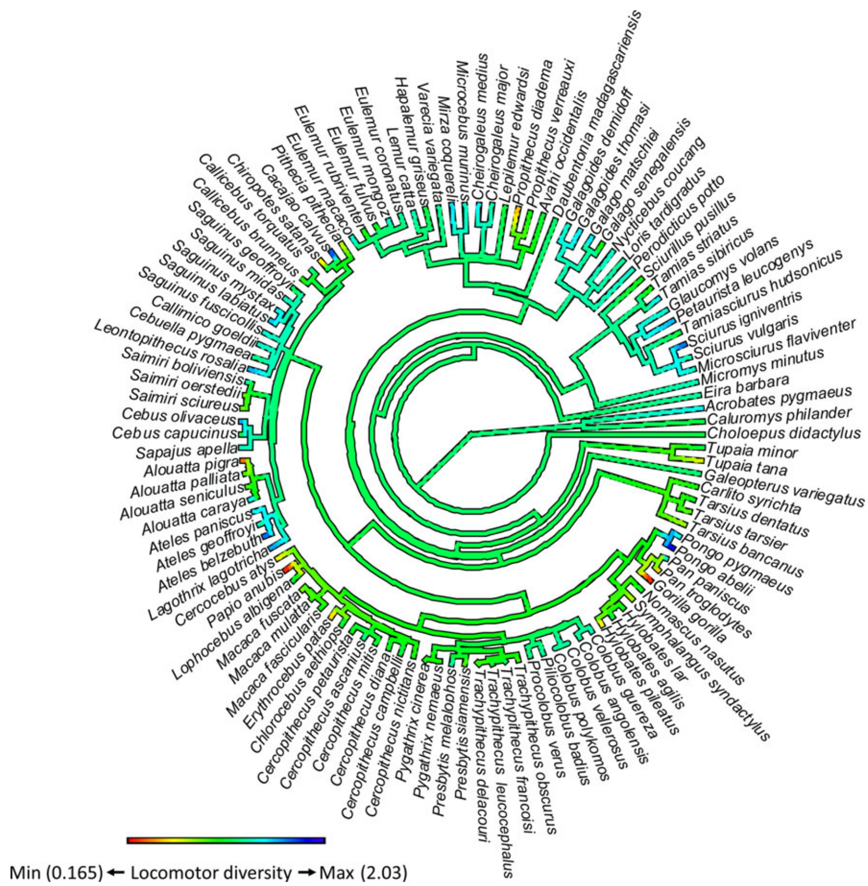

Figure 3 Phylogenetic tree used in this study [extracted and modified from Hedges *et al.* (2015)] with locomotor diversity index mapped onto it using the contMap function of the package phytools. [Colour figure can be viewed at zslpublications.onlinelibrary.wiley.com]

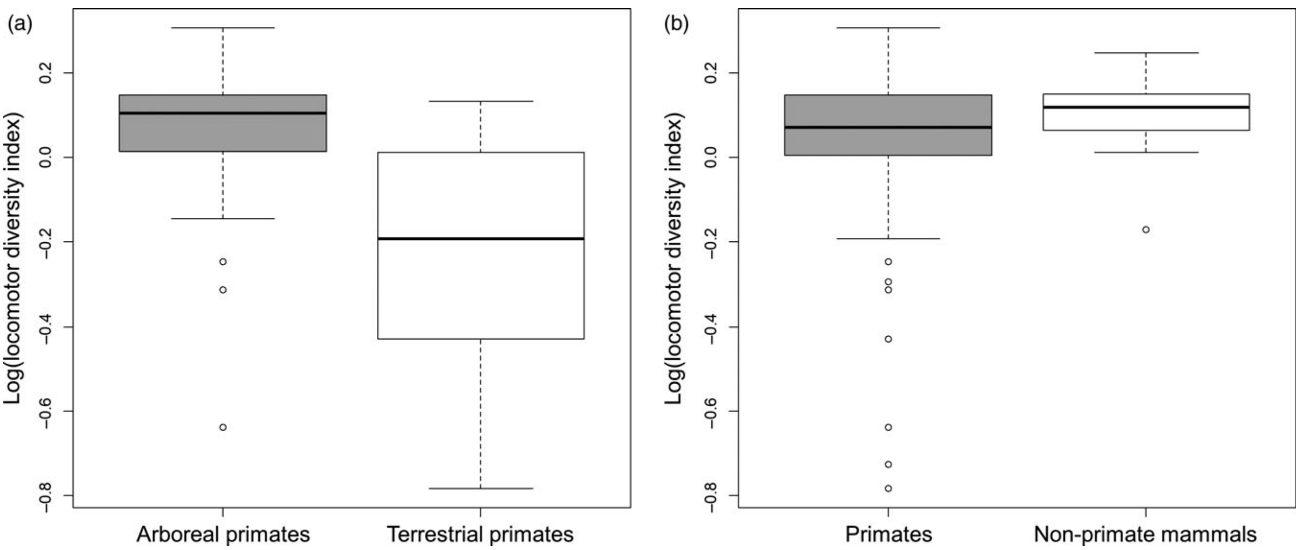

Figure 4 Box and whisker plot demonstrating the difference in locomotor diversity index between (a) arboreal ($n = 84$ species; gray) and terrestrial primates ($n = 9$ species; white) and (b) primate ($n = 93$ species; gray) and non-primate mammals ($n = 17$ species; white). All locomotor diversity indices are logarithmically transformed. The solid black line represents median, the box represents the lower and upper quartiles (25 and 75%), and the whiskers are the minimum and maximum values. Points outside the whiskers represent outliers in the data.

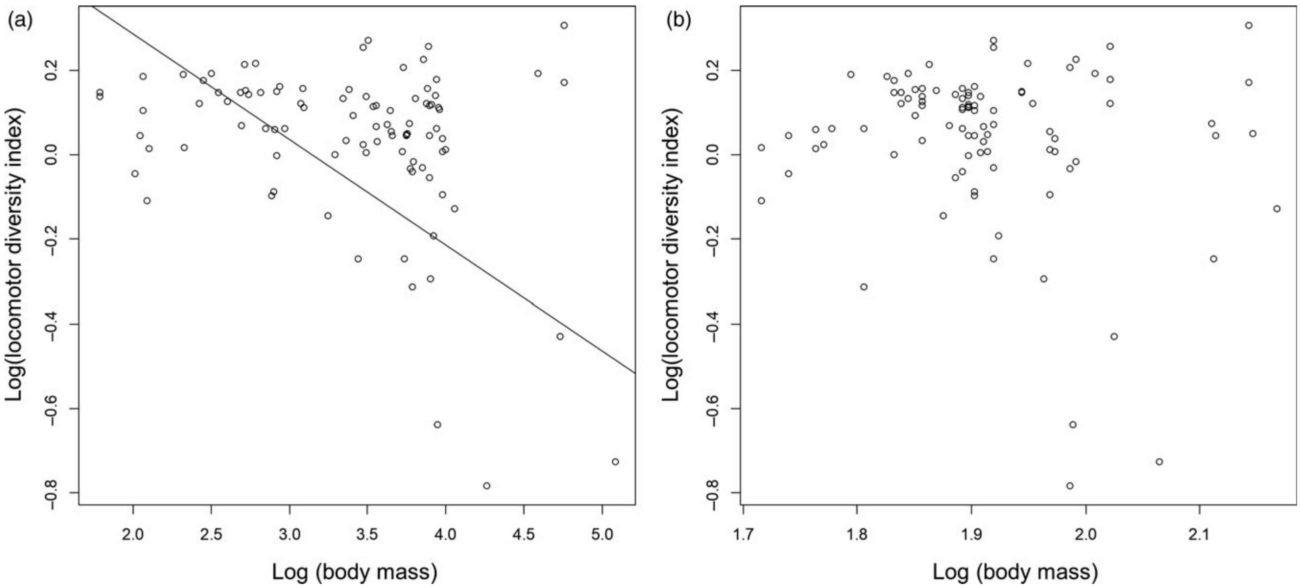

Figure 5 Phylogenetic least squares regression in primates (n = 93 species) demonstrating the (a) significant (P = 0.008) negative ($y = -25x + 0.78$; R^2 = 0.085) relationship between body mass and locomotor diversity index and (b) lack of relationship (P = 0.460) between intermembral index and locomotor diversity index. All data are logarithmically transformed.

species) compared to terrestrial (9 species) primates. As future studies exploring locomotor diversity across mammals are conducted, this hypothesis should be reevaluated.

As initially proposed by Cant (1992), it is likely that body mass variation affects the possible solutions an animal can use when negotiating an arboreal milieu. For example, when crossing a gap body mass may act as a constraint, prohibiting certain solutions (e.g. orangutans cannot cross gaps by leaping), but it may facilitate other solutions not possible by smaller species (e.g. tree-swaying by orangutans). Furthermore, primates are well-known for their use of thin compliant branches. While larger species tend to adopt below branch movements (e.g. bimanual suspension) on these supports (Granatosky, Tripp & Schmitt, 2016), smaller species are less limited in the potential locomotor modes and may adopt above and below branch quadrupedalism, bounding or leaping. Data from this study support the hypothesis that larger species of primate tend to demonstrate lower locomotor diversity than smaller species. However, it is impossible to escape the fact that many large-bodied primates (e.g. *Gorilla gorilla*, *Pan troglodytes* and *Papio anubis*) have essentially abandoned their arboreal habits and have adopted a primarily terrestrial lifestyle. As discussed above, substrate-use appears to be the primary determinant of locomotor diversity in primates. Indeed, when substrate-use is taken into account all effects of body size on locomotor diversity index disappear. This finding likely indicates that body size has a greater effect on whether an animal adopts arboreal or terrestrial substrates rather than constrains an animal's ability to utilize particular locomotor behaviors.

This study observed no relationship between locomotor diversity index and IMI within primates. This means that contrary to the original hypothesis, an anatomically generalized primate is not more likely to have greater locomotor diversity than one with anatomically specializations. This finding is not particularly surprising, as a number of works have demonstrated the ability of primates to easily alter intrinsic aspects gait (e.g. footfall sequence, limb loading, velocity, etc.) depending on different conditions regardless of anatomical specializations (Vereecke, D'Août & Aerts, 2006; Stevens, 2008; Granatosky *et al.*, 2016). With this in mind, it seems improbable that primates would not be able to extend this flexibility to the use of differing locomotor behaviors depending on environmental demands. This finding is especially important for paleontologists, who should be cognizant of the idea that anatomy represents only the minimum set of behaviors that a species is capable of performing. It should be noted, however, that the use of IMI as a proxy for anatomical specialization in primates in this study may have been limiting. Additional studies using other proxies for anatomical specialization should be used to test this finding.

Conclusions

This study presents a quantifiable and repeatable measure of locomotor diversity that is less subject to the biases of the original ethogram or the overemphasis of rare behaviors than simply using raw counts. From these locomotor diversity indices, this study determined that primates do not inherently have greater locomotor diversity than other mammalian taxa. Instead, it is likely that all arboreal mammals, regardless of taxonomic affiliation, must demonstrate high levels of locomotor diversity and flexibility in order to deal with the complexities of moving in a complex three-dimensional environment.

This idea is further emphasized by the ability of primates to adopt a broad range of locomotor behaviors independent of anatomy or body mass. This study represents only a cursory look at locomotor diversity across mammals, and future works focused on quantifying the positional behavior of non-primate mammals are sorely needed to further investigate this topic.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. Locomotor profiles for the species used in this study.
Data S1. Supplementary references.